

NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Target Role: Robotics & Controls Engineer Intern • Ather Energy

+91 6305761091 • saratchandranallabati@gmail.com • Portfolio: saratchandranallabati.github.io • GitHub: github.com/sarat-2007 • LinkedIn

EDUCATION & NATIONAL MERIT PEDIGREE

Amrita Vishwa Vidyapeetham

August 2024 – Expected May 2028

B.Tech in Robotics & Artificial Intelligence (Cumulative GPA: 9.05 / 10.00, Top 5% Cohort)

Amritapuri, India

- **Academic Standing:** AP POLYCET State Merit Rank: 497 (Top 0.5% Statewide Percentile).
- **National Distinctions:** ISRO-IIRS YUVIKA (Selected for space science analytics) | NAEST National Physics Test (Cleared Prelims & Finals, IAPT / Prof. H.C. Verma) | MTSO Olympiad (Level 2 Qualifier).

PEER-REVIEWED PUBLICATIONS & PROCEEDINGS

- N. V. D. S. Sarat Chandra, M. Nunna, Asha C. A., P. C. Sreekumar, “Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity.” in *Proc. IEEE ICRM*, 2025.
- N. V. D. S. Sarat Chandra, et al., “Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation.” Submitted to *IEEE IES*, 2026.
- N. V. D. S. Sarat Chandra, et al., “Domain-Aware Multi-Modal Industrial Diagnostics: Integrating Mechanical Heuristics with Conformal Uncertainty.” Submitted 2026.

TARGETED ENGINEERING PROJECTS & SYSTEMS ARCHITECTURE

Evolution Edge: Self-Evolving Heterogeneous Neural Bridge

Team Lead, AMD Developer Hackathon

AMD Ryzen AI NPU, Vitis AI EP, ONNX Runtime, ROCm 6.x, INT8 PTQ, PEFT/LoRA

- **Core Impact:** Sub-28.4ms deterministic inference @ 8W TDP; offloaded tensor compute across integrated GPUs/NPUs.

Industrial GSM & 4G LTE Cellular IoT Gateway with Dual-Mode Failover

Embedded Firmware Engineer

Valetron A7670C 4G LTE Cat-1, ESP32, FreeRTOS, Circular DMA UART, 240V Relays

- **Core Impact:** <1.8s automatic network failover (WiFi <-> 4G LTE); opto-isolated 240V relay switching with RC snubbers.

Autonomous Mobile Robot (AMR) Edge Platform (IEEE ICRM 2025)

First Author & Lead Developer

ARM Cortex-A, OpenCV, Probabilistic Hough Transforms, Asynchronous UDP Sockets

- **Core Impact:** 35+ FPS lane boundary tracking; <15ms perception-to-actuation telemetry latency.

Pasupatha Autonomous Companion Robot Control Bus

Head of Embedded Systems, Club Mirai

ROS 2 Jazzy, STM32, ESP32, CAN 2.0B, C++ DLS Inverse Kinematics, HIL Testbed

- **Core Impact:** 50 Hz closed-loop control; <50us 5-DOF IK compute cycles (<0.4 deg repeatability).

TECHNICAL SKILLS & CORE COMPETENCIES

Languages & Runtimes: C (C99/C11), C++ (C++17/20), Python (3.12), Bash, MATLAB, POSIX Sockets, FreeRTOS, Bare-Metal

Robotics, AI & Middleware: ROS 2 (Jazzy), CycloneDDS, Forward/Inverse Kinematics, OpenCV, PyTorch, PINNs, ONNX Runtime, Vitis AI EP

Hardware, Buses & Silicon: CAN Bus (2.0B), UART DMA, I2C (9-clock recovery), SPI, Valetron A7670C (4G LTE), STM32, ESP32, Raspberry Pi 5

Simulation & Tooling: ANSYS Fluent (CFD), SolidWorks, Fusion 360, Linux (Ubuntu Server), Docker, Git/GitHub, LaTeX